

EOD Report – Jenkins Pipelines

Date: 07 October 2026

Task: Jenkins Pipelines – CI/CD Implementation

Source Code: https://github.com/Zeeshancloud15/Tomcat_app01.git

1. Objective

Worked on Jenkins Pipelines using the **Tomcat_app01** application repository and implemented different Jenkins Pipeline types to understand CI/CD workflow and Pipeline syntax.

2. Tasks Completed

1. Declarative Pipeline Job

Created:

tomcat-app01-declarative

Implemented stages:

Checkout

↓

Build

↓

Test

↓

Package

↓

Deploy

Used Declarative Pipeline syntax:

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                // checkout code
            }
        }

        stage('Build') {
            steps {
                sh 'mvn clean compile'
            }
        }

        stage('Test') {
            steps {
                sh 'mvn test'
            }
        }
    }
}
```

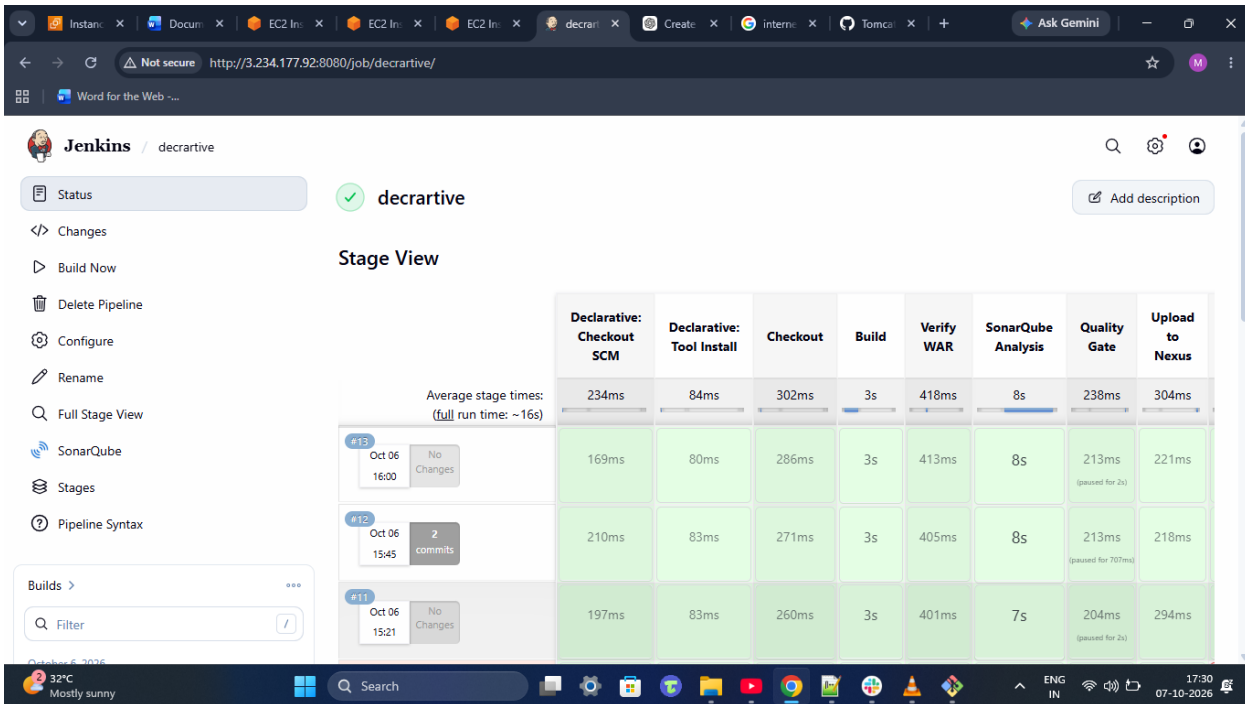
```

    }
}

stage('Package') {
    steps {
        sh 'mvn package -DskipTests'
    }
}

stage('Deploy') {
    steps {
        echo 'Deploy application'
    }
}
}

```



2. Scripted Pipeline Job

Created:

tomcat-app01-scripted

Implemented the CI/CD stages using Scripted Pipeline syntax:

Checkout

↓

Build

↓

Test

↓

Package

↓
Deploy
Used:
node {

```
    stage('Checkout') {  
        // Git checkout  
    }  
  
    stage('Build') {  
        sh 'mvn clean compile'  
    }  
  
    stage('Test') {  
        sh 'mvn test'  
    }  
  
    stage('Package') {  
        sh 'mvn package -DskipTests'  
    }  
  
    stage('Deploy') {  
        echo 'Deploy application'  
    }  
}
```

The screenshot shows the Jenkins web interface for a pipeline named 'scripted-pipeline-job'. The 'Stage View' is displayed, showing a table of stage execution times for two builds. The first build, #65, completed successfully on Oct 06 at 16:57 with 2 commits. The second build, #66, is in progress on Oct 06 at 16:56 with no changes. The stages and their durations are:

Stage	Source Checkout	Compile and Package	Artifact Validation	Code Quality Analysis	Quality Validation	Publish Artifact	Application Deployment	Notify Success
Average stage times (full run time: ~15s)	199ms	3s	296ms	7s	94ms	101ms	77ms	103ms
#65 Oct 06 16:57 2 commits	199ms	3s	296ms	7s	94ms (passed for 2s)	101ms	77ms	103ms
#66 Oct 06 16:56 No Changes								

Below the stage view, the 'SonarQube Quality Gate' section is visible, showing a green status for the current build.

3. Multi-Stage Pipeline Job

Created:

tomcat-app01-multistage

Implemented a complete sequential CI/CD flow:

GitHub



Checkout



Build



Test



Package WAR

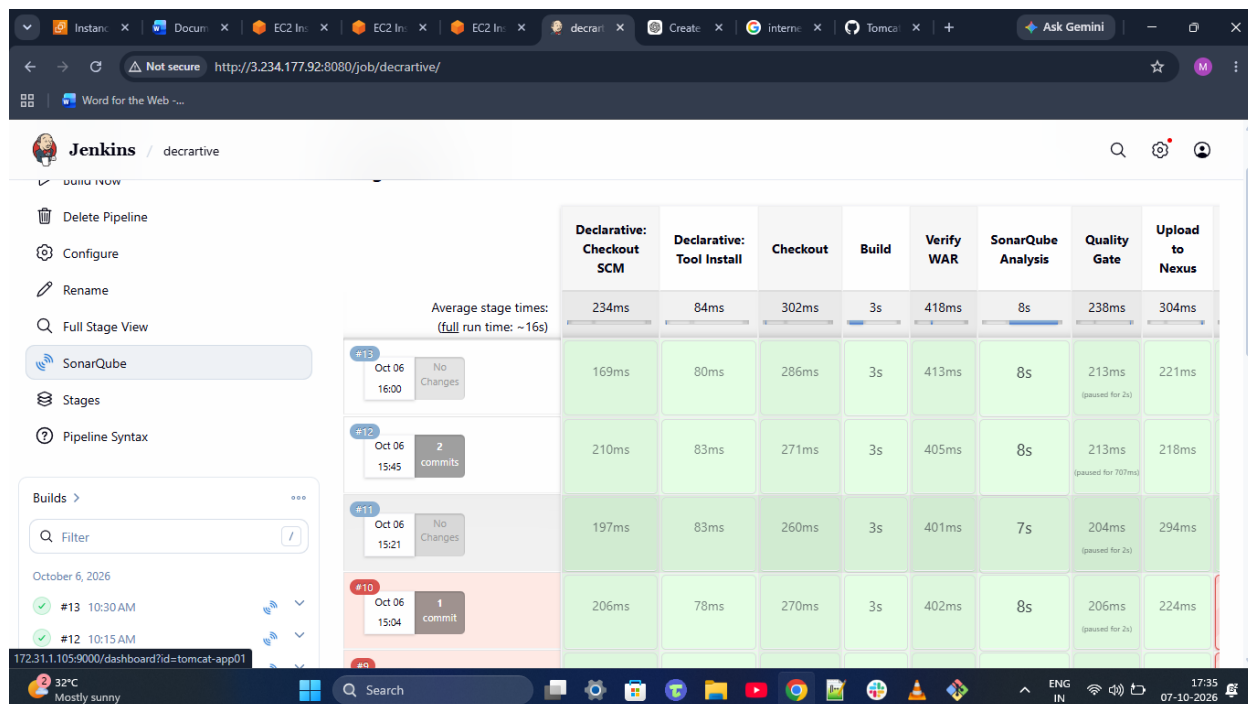


Deploy



Verify

The pipeline separates each CI/CD activity into an individual Jenkins stage, making the execution easy to monitor through Jenkins Stage View.



4. Parallel Stage Pipeline Job

Created:

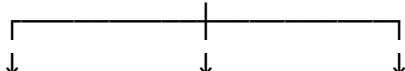
tomcat-app01-parallel

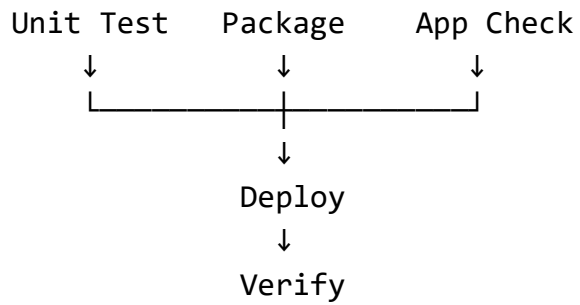
Implemented parallel execution for independent checks.

Checkout



Build





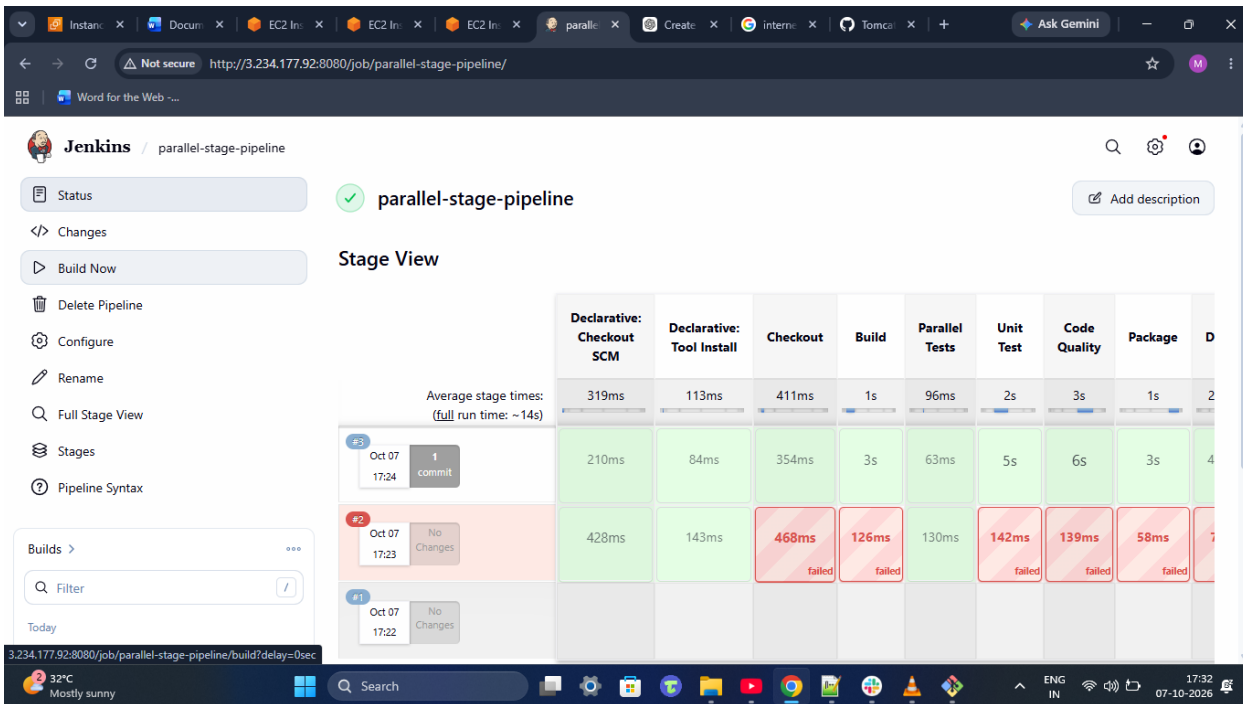
Used Jenkins Declarative parallel syntax:
stage('Parallel Checks') {

```
    parallel {

        stage('Unit Test') {
            steps {
                sh 'mvn test'
            }
        }

        stage('Package Check') {
            steps {
                sh 'mvn package -DskipTests'
            }
        }

        stage('Application Check') {
            steps {
                sh 'ls -la'
            }
        }
    }
}
```



3. Jenkins Tools Used

- Jenkins
- Git/GitHub
- Maven
- Java
- Jenkins Pipeline
- Declarative Pipeline
- Scripted Pipeline
- Parallel stages
- Tomcat deployment concept

Maven installation configured in Jenkins:
maven-integration

4. Key Learning

- **Declarative Pipeline:** Uses a structured pipeline `{}` syntax and is easier to read and maintain.
- **Scripted Pipeline:** Uses Groovy-based syntax and provides more flexibility for complex logic.
- **Multi-Stage Pipeline:** Divides CI/CD into multiple sequential stages.
- **Parallel Pipeline:** Runs independent stages simultaneously to reduce execution time.
- **Jenkinsfile:** Defines the CI/CD pipeline as code and can be stored in GitHub.
- **Stage View:** Helps identify the status and failure point of individual pipeline stages.

5. Issues / Checks

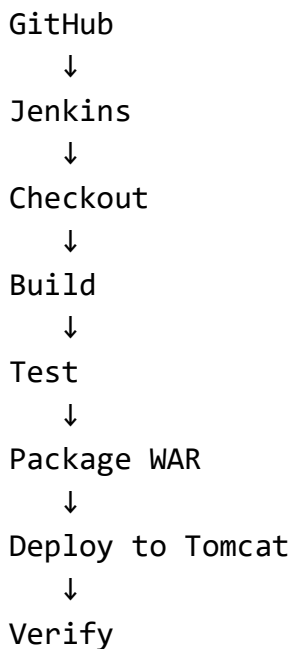
- Verified Maven configuration and used the Jenkins Maven installation name maven-integration.
- Ensured Java, Maven and Git are available on the Jenkins agent.
- Reviewed Git repository and configured the Tomcat_app01 source repository.

- Deployment stage was kept configurable because the actual Tomcat server/credential details were not specified.

6. Final Status

Task	Status
Declarative Pipeline	✓ Completed
Scripted Pipeline	✓ Completed
Multi-Stage Pipeline	✓ Completed
Parallel Stage Pipeline	✓ Completed
GitHub Integration	✓ Configured
Maven Build	✓ Configured
Unit Testing	✓ Configured
WAR Packaging	✓ Configured
Tomcat Deployment Stage	✓ Prepared

Final CI/CD Flow



Overall Status: Completed successfully.

Available next action: Create a downloadable PDF file here in this chat containing the finalized decisions and immediate actions above